Predicting 30-Day Hospital Readmission

Task: predict whether a patient is readmitted within 30 days of discharge

Metric: ROC-AUC on predicted probabilities

Github repository: eyk-1/ml-project-runs

1. Overview

The task is to predict the probability that a hospital stay is followed by a readmission within 30 days *(readmit_30d)*. The training set contains **76,975** stays and the test set contains **32,812**. About **10.1%** of the training labels are positive. Kaggle scores submitted probabilities using ROC-AUC.

The data was joined and aggregated such as the linked encounter, patient, hospital, diagnosis, procedure, laboratory, vital-sign, medication, unit-transfer, discharge-planning and prior-utilization data. All the inconsistent units and codes were cleaned up, informative missingness was kept, and trained models on a common five-fold stratified split.

A total of twelve submissions were made. Final leaderboard score achieved is the private score of **v10**, **0.75657 (public 0.74752)**: Kaggle scores the selected submission with the best private score, and we selected **v9, v10 and v12**. The best public score was **v9 at 0.74755**. The corrected **LightGBM** and **EBM** rank blend behind **v9** and **v10** had a five-fold out-of-fold (OOF) AUC of about 0.7526.

The private leaderboard uses the 70% of test stays that the public leaderboard did not. Every submission from v4 onwards scored between **0.75501** and **0.75663** privately, so the candidates compared are too close to separate reliably. Regularised gradient boosting was the most useful approach, and differences of a few ten-thousandths in OOF AUC did not predict the public or private ordering.

2. Data pipeline

2.1 Tables used and dropped

All 11 encounter and patient data tables were used: **index_encounters, patients, hospitals, diagnoses, procedures, lab_results, vitals, medications, unit_transfers, discharge_planning and prior_utilization**. We used *icd10_lookup* to add diagnosis chapter and chronic-condition information. **data_dictionary.csv** was reference material; **train.csv** and **test.csv** supplied the split and labels. We dropped no source table.

2.2 Joins

index_encounters is the encounter spine. Patient and hospital attributes were joined to it; every many-row table was first reduced to one row per **encounter_id**, then left-joined. Prior-utilization visits were restricted to dates before the index admission. The label was attached only for training. Both splits used the same row-level cleaning and feature construction.

2.3 Aggregation of many-row tables

Table	Rows / coverage	Features aggregated per stay
diagnoses	**451,976 / 100%**	*Counts; primary code and chapter; chapter counts; common-code indicators; chronic burden and Charlson-style score.*
procedures	**19,547 / 17.1%**	*Count, first code, distinct codes, and timing relative to admission and discharge.*
lab_results	**1,668,783 / 100%**	*Per-test last, mean, min, max, count, first-to-last change, discharge timing, abnormal flags and trends.*
vitals	**2,267,367 / 100%**	*Per-vital last, mean, min, max, standard deviation, count and change; discharge-window summaries and abnormal counts.*
medications	**447,278 / 98.0%**	*Counts, distinct and common drug names, continuation, route, drug class, and start/stop timing.*
unit_transfers	**211,097 / 100%**	*Number and type of transfers, hours per unit, ICU use and discharge-relative timing.*
discharge_planning	**97,552 / 88.9%**	*Fields retained, plus row-present and missing-follow-up indicators.*
prior_utilization	**714,062 / 99.5% of patients**	*ED, inpatient and outpatient counts by lookback window; time since prior visits; same- versus other-hospital use.*

Table 1. Core multi-row tables and their encounter-level summaries.

3. Final dataset

The final feature table contains **109,787** rows: **76,975** training and **32,812** test stays. It stores **461** predictors (**442 numeric and 19 categorical**). Four additional fold-local target encodings are generated during model fitting. Two high-cardinality raw columns, **attending_id** and **proc_first**, are then dropped, giving **463** model inputs. The target and **encounter_id** are not model predictors.

The complete feature inventory is attached as experiments/final_columns_v4.csv. It lists each of 465 stored or model-time columns, with its source, construction, type and percentage missing.

Feature block	Columns	Notes
Version 1	297	*Core joins and encounter-level aggregates.*
Version 2	17	*Clinical burden scores, ratios and data-quality indicators.*
Version 3	145	*Discharge state and timing features for labs, vitals, medications, procedures, transfers and prior visits.*
Version 4	2	*Final corrected pipeline additions.*
Fold-local target encodings	4	*Hospital, attending, primary diagnosis and first procedure; computed within each training fold.*
Inventory total	**465**	*461 stored predictors plus four model-time encodings.*

Table 2. Feature inventory by the version in which each block first appeared. The two high-cardinality raw identifiers are documented but excluded from the fitted model matrix.

The model input is assembled inside each fold. Category levels, target encodings and hospital and attending counts are learned from that fold's training rows only; unseen validation or test categories receive the appropriate missing or global fallback value.

4. Data quality

The source tables contain inconsistent formats, duplicate records and implausible measurements. The same corrections are applied to training and test data.

Problem	Evidence	Handling
Exact duplicates	*Diagnoses 902; labs 3,331; vitals 2,265; medications 893; prior visits 1,284.*	*Exact duplicates removed before aggregation.*
Mixed laboratory units	*Albumin, creatinine, glucose and hemoglobin appear in mixed units.*	*Convert to a common unit per test before summarising.*
Implausible laboratory results	*5,194 results (0.31%) outside physiological ranges after conversion.*	*Set to missing; counts and measurement summaries retain information about data quality.*
Temperature units	*66,538 of 378,097 readings (17.6%) exceed 45 degrees.*	*Convert Fahrenheit values to Celsius.*
Impossible vital signs	*369 readings, including SpO2 of 0 and heart rate of 300.*	*Set impossible readings to missing.*
Malformed ICD-10 codes	*182,326 codes lack the decimal; 124,211 rows initially miss the lookup.*	*Insert the decimal after the third character and join the lookup.*
Drug-name variants	*122 spellings for 52 drugs.*	*Normalise case and whitespace before building drug features.*
Procedure systems	*CPT and CPT-4 both appear.*	*Treat them as one code system.*
Placeholder categories	*?, U, Not Available and Not Mapped occur in categorical fields.*	*Convert placeholders to missing; standardise insurance-type case.*
Implausible dates and values	*Very early birth years, zero or very long stays, remote follow-up dates.*	*Set invalid values to missing; parse each table's date format explicitly.*

Table 3. Data-quality issues and corrections.

5. Missing values

Missingness often reflects clinical decisions, for example specialised tests are ordered for patients who need them. The explanations below are interpretations of the dataset, not observed causes.

Where	Missingness / pattern	Interpretation and response
Admission weight	*87.0% missing; readmission 12.6% when recorded vs 9.7% otherwise.*	*Recorded for a subset of stays; preserve missingness and an explicit flag.*
Attending specialty	*46.4% missing; readmission rates are similar.*	*Likely source-system gaps; retain a missing category and flag.*
Procedures	*No row for 82.9% of stays.*	*Usually no procedure; absent procedure counts become zero, timing stays missing.*
Discharge planning	*No row for 11.1%; follow-up days missing in 41.8% of existing rows.*	*May mean planning was not done or not recorded; add row-present and missing-follow-up flags.*
Selective labs	*Albumin ordered in 37.2%; HbA1c 16.0%; troponin 14.4%; BNP 12.3%; lactate 9.7%.*	*Test ordering is informative; keep counts and missing values.*
Registration fields	*Insurance 17.2%; marital status 9.0%; language 7.1%; deprivation 4.7%; ZIP 4.0%.*	*Retain missing categories; tree models can use missingness directly.*
Prior visits	*No prior-utilization rows for 0.5% of patients.*	*Represents first-time patients; counts become zero and timing remains missing.*

Table 4. Main missing-data patterns. Rates are from the training data.

- **LightGBM** and **other boosted-tree models** handle numeric missing values natively; explicit missing indicators are added when useful.
- Imputation for models that require complete matrices is fitted inside each training fold, with missing indicators and a separate missing category for text.
- Target encodings use an inner five-fold scheme on the outer training partition, so a training row does not receive an encoding derived from its own label.

6. Models

6.1 How we tuned and evaluated

- Models use **StratifiedKFold** with five folds, *shuffle=True* and **random_state=0** on the **76,975** training stays.
- Preprocessing, target encodings, category alignment and provider counts are fitted from each outer training fold. **LightGBM** iteration counts use early stopping on that fold's validation portion.
- The main **LightGBM** setting is strongly regularised: four leaves, at least **400 rows per leaf, 20% feature sampling, L2 = 60 and learning rate 0.04**. Target-encoding smoothing was screened across **M = 10, 25, 50, 100 and 200**.
- Blends combine OOF predictions by rank, then map ranks back to a **LightGBM** probability scale. Their weights were screened on **80%** of OOF rows. This is a useful diagnostic, but because the underlying OOF folds differ from that 80/20 split, it is not a fully nested estimate.
- Resampling and re-weighting was not applied to the classes in the main models. ROC-AUC depends only on how patients are ranked.

6.2 Three strongest individual components

Model	Key hyperparameters	Five-fold OOF AUC
Corrected LightGBM	*4 leaves; min leaf 400; feature fraction 0.2; L2 = 60; learning rate 0.04; M = 50 target encoding; early stopping.*	**0.752053**
LightGBM boosted-stump GAM	*2 leaves; depth 1; learning rate 0.1; min child 100; feature fraction 0.5; L2 = 20; max_bin 63.*	**0.751929**
Corrected EBM	*150 LightGBM gain-selected features per fold; 10 pairwise interactions; two outer bags.*	**0.750382**

Table 5. Best standalone components. None of these three was submitted alone; the scores in Section 6.4 are for their ensembles.

6.3 Other model and feature experiments

Experiment	OOF AUC	Outcome
LightGBM with version 3 discharge-state features	**0.7518**	*Below the corrected version 2 setting.*
LightGBM with monotone constraints	**0.752043**	*Effectively tied with the baseline; blend gain was negligible.*
Positive-class weight 0.75	**0.751981**	*Below the unweighted LightGBM; a small blend gain did not transfer to the public score.*
Clinical interaction features	**0.751675**	*Lower than baseline.*
Hospital x primary-diagnosis target encoding	**0.7514**	*Did not improve the baseline.*
Hospital x admission-month target encoding	**0.751576**	*Did not improve the baseline.*
Hospital x service-line target encoding	**0.751616**	*Did not improve the baseline.*
Extra early/late lab and vital windows	**0.751442**	*Existing features already included lab trends and discharge summaries; added windows did not help.*
XGBoost DART	**0.746786**	*Weaker than the standard LightGBM.*
AdaBoost	**0.7378**	*Weaker than the main boosted-tree models.*
Logistic regression with pairwise numeric interactions	**0.7337**	*No useful gain.*

Table 6. Selected corrected-pipeline model and feature tests. Scores use the same five OOF folds unless otherwise stated.

Other algorithms/models evaluated (but not submitted) include: **CatBoost**, **XGBoost**, **spline logistic regression**, **random forests**, **ExtraTrees**, **SVM**, **Naive Bayes**, **KNN**, **HistGradientBoosting** and **LambdaRank**. These did not produce a stronger public submission. **Diagnosis** and **medication SVD** features and a large set of explicit pairwise features were also neutral or harmful. A patient-ID target encoding was not pursued: each encounter has a unique **patient_id**, and there is no patient overlap between train and test.

Experiment log: (also on github):

ID	Model	Key params	Public LB
v1	***LightGBM, 3-seed avg***	*lr 0.03, 15 leaves, 290 rounds*	**0.74326**
v2	***LightGBM regularised + v2 features + target encoding***	*lr 0.04, 4 leaves, 700 rounds*	**0.74589**
v3	***CatBoost, 2-seed avg, native categoricals***	*depth 5, lr 0.06, 445 rounds*	**0.74337**
v4	***Rank blend: 0.8 LightGBM (v2) + 0.2 Logistic Regression***	*LR C = 0.02*	**0.74682**
v5	***XGBoost, 3-seed avg, target encoding***	*eta 0.04, depth 3, 363 rounds*	**0.74397**
v6	***Rank blend: 0.4 LightGBM (v2 features) + 0.4 LightGBM (v3 features) + 0.2 Logistic Regression***	*weights from OOF grid search*	**0.74685**
v7	***Kaggle notebook: rank blend 0.4 LightGBM (v2 features, 3 leaves) + 0.4 LightGBM (v3 features, 3 leaves) + 0.2 Logistic Regression***	*settings chosen by CV inside the notebook*	**0.74679**
v8	***Rank blend: LightGBM v2 + v3 + spline logistic regression + EBM***	*weights 0.3 / 0.3 / 0.2 / 0.2*	**0.74695**
v9	***Rank blend: corrected LightGBM + corrected EBM (clean fold-local pipeline, features v4)***	*weights 0.65 / 0.35; CV 0.7526*	**0.74755**
v10	***Rank blend: corrected LightGBM (5-seed final fit) + corrected EBM***	*weights 0.65 / 0.35; CV 0.7526*	**0.74752**
v11	***Rank blend: corrected v9 blend + boosted-stump GAM***	*GAM weight 0.325; CV 0.752794; holdout 0.741811*	**0.74714**
v12	***Rank blend: corrected v9 + LightGBM with positive-class weight 0.75***	*17.5% added rank weight; CV 0.752600; holdout 0.741773*	**0.74728**

6.4 Ensembles and scored submissions

The four best public submissions are compared below. v9 had the highest public score and v10 the highest private score among the three submissions we selected. v11 had the highest OOF AUC, but neither its public nor its private score was the best.

ID	Ensemble	OOF AUC	Public AUC	Private AUC	Selected
v9	**Corrected LightGBM + EBM; rank weights 0.65 / 0.35.**	**0.752575**	**0.74755**	**0.75598**	***Yes***
v10	**Corrected LightGBM with five-seed final fit + EBM; rank weights 0.65 / 0.35.**	**0.7526**	**0.74752**	**0.75657**	***Yes***
v11	**v9 blend + boosted-stump GAM; 32.5% GAM rank weight.**	**0.752794**	**0.74714**	**0.75624**	***No***
v12	**v9 blend + class-weight-0.75 LightGBM; 17.5% added rank weight.**	**0.752600**	**0.74728**	**0.75601**	***Yes***

Table 7. Best four public submissions with their OOF, public and private AUC.

The **v9** and **v10** public scores differ by **0.00003**, and their private scores differ by **0.00059**. The remaining experimental blends changed the reserved OOF holdout by less than **0.0001**, which is too small to trust. Rank averaging suits ROC-AUC because it evaluates ordering rather than probability calibration. Each submission still contains probabilities, obtained by mapping the blend ranks back to the corrected **LightGBM** scale.

6.5 Why five-fold cross-validation, and how well it held up

Five-fold stratified cross-validation was used because of four reasons. Each fold holds about **1,550 positive stays (7,755 overall)**, which is enough for a stable AUC. Each model trains on **80%** of the data, close to the final fit on all of it. Using the same folds for every model let us blend *out-of-fold* predictions. And ten folds or repeated folds would have cost two to three times the computing on our hardware.

It estimated the level of performance well. Averaging each submission's public and private AUC with **30%** and **70% weights**, which approximates its score on the whole test set, gives a value **0.0006** to **0.0020** above its OOF AUC (for **v10**, about **0.7539** against **0.7526**). The estimate is slightly conservative because the final models train on 100% of the training data and not **80%**. What it could not do is rank near-identical candidates: **fold-to-fold** AUC varies by about **0.008**, so differences below **0.001** between blends are within noise. Repeating the five-fold split with several seeds would have measured that noise directly.

7. Results and discussion

7.1 Leaderboard result

The final leaderboard score is the private score of **v10, 0.75657**. Its **public scor**e was **0.74752**. The best public score achieved was **v9** at **0.74755**. Of the three submissions selected (**v9, v10 and v12**), **v10** had the best private score, so it is the one Kaggle uses.

Private scores were between **0.0084** and **0.0110** higher than public scores for every submission, and the public ordering did not carry over: **v8** had the highest private score of all (**0.75663, not selected, 0.00006 above v10**), while **v9** had the highest public score. The gap comes from the two test samples, not from any one model.

7.2 Which approach performed best and why

The **corrected LightGBM** and **EBM rank blend (v9, v10)** is the best submission. **LightGBM** provides the strongest standalone ranking, handling mixed numeric and categorical features and missing values with shallow, heavily regularised trees. **EBM** adds a different additive structure. The **0.65 / 0.35** blend reached OOF AUC of **0.752575**.

The result did not improve simply by adding more models. The **stump-GAM blend** had the **highest** OOF score (**0.752794**) but a lower public score (**0.74714**), and the class-weighted blend reached OOF **0.752600** but scored **0.74728**.

7.3 Why the score may be difficult to raise

- Several different model families and feature sets land near the same OOF range, around **0.75**.
- The strongest model and blend differences are only **a few ten-thousandths of AUC**, and the public and private scores did not follow those tiny CV gains. Hospital, diagnosis, utilization and clinical features are all represented, and the main data-quality corrections and temporal summaries are already included.

These results suggest the remaining gap is hard to capture with the available data, though they do not prove a better model is impossible.

7.4 Limitations

- Early stopping uses each validation fold to select the number of boosting rounds, giving a small optimism to the reported OOF scores.
- Blend weights were screened on a subset of OOF predictions, but the base OOF folds were not nested around that subset. This is not a fully independent estimate of the whole blend-selection process.
- All comparisons use one **five-fold** split. Differences under **0.001** should not be over-interpreted, and scores vary by split and run.

7.5 What we would try next

- Evaluate a logistic or non-negative meta-learner using fully nested stacking: for each untouched outer fold, generate fresh inner-fold predictions using only the outer training data, train the meta-learner on those predictions, then score the outer fold.
- Repeat the main comparison over several outer split seeds to estimate how stable the small score differences are.

Appendix A. Submission history

All twelve submissions with their recorded five-fold OOF, public and private AUC. The final leaderboard score uses the best private score among the selected submissions.

ID	Model / blend	OOF AUC	Public AUC	Private AUC	Selected
v1	**LightGBM, 3 seeds, 15 leaves.**	**0.7494**	**0.74326**	**0.75402**	**-**
v2	**Regularised LightGBM, version 2 features, target encoding.**	**0.7521**	**0.74589**	**0.75557**	**-**
v3	**CatBoost, 2 seeds.**	**0.7476**	**0.74337**	**0.75231**	**-**
v4	**0.8 LightGBM + 0.2 logistic regression.**	**0.7525**	**0.74682**	**0.75626**	**-**
v5	**XGBoost, 3 seeds, target encoding.**	**0.7502**	**0.74397**	**0.75501**	**-**
v6	**0.4 LightGBM v2 + 0.4 LightGBM v3 + 0.2 logistic regression.**	**0.7527**	**0.74685**	**0.75624**	**-**
v7	**Kaggle notebook blend: two LightGBM feature sets + logistic regression.**	**0.7528**	**0.74679**	**0.75635**	**-**
v8	**LightGBM v2/v3 + spline logistic regression + EBM.**	**0.7528**	**0.74695**	**0.75663**	**-**
v9	**Corrected LightGBM + EBM rank blend.**	**0.752575**	**0.74755**	**0.75598**	***Yes***
v10	**Five-seed corrected LightGBM + EBM rank blend.**	**0.7526**	**0.74752**	**0.75657**	***Yes***
v11	**v9 + boosted-stump GAM rank blend.**	**0.752794**	**0.74714**	**0.75624**	**-**
v12	**v9 + class-weight-0.75 LightGBM rank blend.**	**0.752600**	**0.74728**	**0.75601**	***Yes***

Table 8. Kaggle submission history. The highest public score was v9 (0.74755); the highest private score among selected submissions was v10 (0.75657).

Appendix B. Attached files and reproducibility

- **experiments/final_columns_v4.csv**: 465-row inventory of the corrected feature table, including source, construction method, feature version, data type and missingness.
- **submissions/kaggle_readmission_notebook_v2.ipynb**: notebook for the corrected v9 feature and model pipeline. The earlier **kaggle_readmission_notebook.ipynb** records the previous end-to-end version.
- **src/**: feature construction, cross-validation and model scripts. README.md describes the project layout; experiments/results.csv records model and submission scores.
- **Challenge.pdf**: assignment brief.

All experiments use only the competition data. Submission files follow ***sample_submission.csv*** and contain predicted probabilities for every test **encounter_id**. The report and code do not include the raw competition data; the data files remain local and are excluded from **Git**.